Clautolisp Implementation Roadmap

Codex

Purpose

This document is the architectural plan for bringing `clautolisp` from its current Phase-6 state (reader + evaluator + small core builtins, dialect-aware) to broad coverage of the AutoLISP / Visual LISP language family documented in `autolisp-spec`.

It distinguishes:

- **function coverage** — adding pure-language builtins that do not touch a CAD host (numeric, list, string, control, error);
- **host coverage** — adding the host-facing surface that real AutoLISP code depends on (entities, selection sets, tables, sysvars, command dispatch, graphics, dialogs, reactors);
- **backends** — the runtime swappable implementations of the host surface: a deterministic **MockHost**, an optional **GuiHost**, and a future **LiveHost** adapter to BricsCAD or AutoCAD.

The plan is intentionally headless-first: every layer must be testable without a graphical CAD process. A GUI backend is a later opt-in, not a prerequisite.

The plan is intentionally driven by the spec, not by an internal wishlist: each phase below references the chapters of the `autolisp-spec` document that define its scope.

Strategic Position

The architectural choice from chapter 1 of the spec stands:

1. **Implement AutoLISP as a Common-Lisp evaluator, not as a translation into Common Lisp.**
2. **Common Lisp is the implementation and extension language.** It does not provide the semantics.
3. *Host effects go through a single abstract interface (the HAL — Host Abstraction Layer) so that backends can be swapped at session-creation time.*

The dialect descriptor introduced in Phase 6 (`strict`, `autocad-2026`, `bricscad-v26`) controls **language-level** product divergence. The HAL choice is **orthogonal**: a dialect describes what the source code is allowed to mean; a backend describes where its host effects land.

Layered Architecture

```
+----------------------------------------+
|  AutoLISP source (your application)    |
+----------------------------------------+
|         Reader  (autolisp-reader)      |   chapter 2
+----------------------------------------+
|        Evaluator (autolisp-runtime)    |   chapter 3
+----------------------------------------+
|     Builtin registry (-builtins-core) |   chapters 4-7, 9-15
+----------------------------------------+
|  HAL - Host Abstraction Layer (NEW)    |   chapters 16, 18-22
+----------------------------------------+
|   Backend: MockHost | GuiHost (opt)    |
|            LiveHost (deferred)         |
+----------------------------------------+
```

The reader, runtime, and core builtins exist today and are sufficient for any program whose surface is values + control flow. The HAL is the next architectural addition; everything beyond it is backend-specific work.

Data Model — Inferred From the CAD Specifications

Most AutoLISP host functions operate on a small set of recurring data shapes. The runtime needs explicit Common-Lisp structures for each, even when the active backend stores nothing real behind them.

Entity (ENAME)

A handle naming a single CAD object. AutoLISP code obtains an ENAME from `entlast`, `entnext`, `ssname`, etc., and treats it opaquely. It is round-tripped through `entget` (returns a **DXF group-code list** — a list of dotted pairs of group-code integers and values) and `entmod` / `entupd` / `entdel` / `entmake` / `entmakex`.

```
(defstruct entity-handle
  (id      (gensym "ENT-") :type t)   ; opaque key into the backend
  (kind    nil :type symbol)          ; :line / :circle / :insert / ...
  (block   nil)                       ; owning block-table-record handle
```

```
  (layer   nil)                        ; layer-table-record handle
  (data    nil :type list)             ; cached DXF group-code list
  (deleted nil :type boolean))
```

The DXF group-code list (`data`) is the primary AutoLISP-visible representation: `(0 . "LINE") (5 . "1A") (8 . "0") (10 5.0 5.0 0.0) (11 7.0 5.0 0.0)` etc. Every entity-modifying builtin works against that shape; the backend is free to keep additional fields (graphic state, ObjectId, etc.) on the side.

Selection set (PICKSET)

A bag of ENAMEs with insertion-order-preserving identity. Built by `ssget` from filter lists; manipulated by `ssadd`, `ssdel`, `sslength`, `ssname`. `ssgetfirst` / `sssetfirst` are pickfirst hooks.

```
(defstruct pickset
  (id      (gensym "SS-") :type t)
  (members nil :type list))           ; ordered list of entity-handle
```

Tables (block, layer, ltype, style, view, ucs, vport, dimstyle, appid)

Symbol tables in DXF terminology. `tblnext`, `tblsearch`, `tblobjname`. `entget` on a table-record returns the same DXF list shape.

```
(defstruct symbol-table-record
  (id      (gensym "TBL-") :type t)
  (kind    nil :type symbol)          ; :block / :layer / :ltype / ...
  (name    "" :type string)
  (data    nil :type list))
```

The session keeps one **symbol-table** per kind, indexed by name.

Dictionaries

Generic key→object containers anchored at the named-object dictionary. `namedobjdict`, `dictsearch`, `dictnext`, `dictadd`, `dictremove`, `dictrename`. Dictionary entries can be entities, xrecords, or sub-dictionaries.

```
(defstruct dictionary
  (id      (gensym "DICT-") :type t)
  (entries (make-hash-table :test #'equal)))   ; string -> handle
```

Symbol tables for system variables

`getvar` / `setvar` access named state. `initget` / `getstring` / `getpoint` etc. read interactive input. The session needs a sysvar table:

```
(defstruct sysvar-table
  (entries (make-hash-table :test #'equalp)))   ; name -> sysvar-cell

(defstruct sysvar-cell
  (name    "" :type string)
  (kind    nil :type symbol)          ; :integer / :real / :point / :string / :short
  (value   nil)
  (read-only-p nil :type boolean))
```

SAFEARRAY / VARIANT

The Visual LISP COM bridge. SAFEARRAYs are typed multi-dimensional arrays; VARIANTs tag a value with its COM type. Modeled now (the runtime already has `autolisp-safearray`, `autolisp-variant`, `autolisp-vla-object` stubs); operators in the `vlax-*` family will fill them in.

VLA-object

A handle to a COM-exposed object (Layer, Block, Document, etc.). The runtime stores an opaque ID into a per-session table; the backend decides what the ID means.

```
(defstruct vla-object-handle
  (id          (gensym "VLA-") :type t)
  (kind        nil :type symbol)      ; :document / :layer / :block / :line / ...
  (entity-back nil)                   ; ename if applicable
  (released-p  nil :type boolean))
```

Layer states

Persistent named snapshots of layer-property bundles. `layerstate-*` family.

Reactors

Object-, document-, and editor-level callback registrations. Modeled per `vlr-*` entry: a callback list keyed by event class.

DCL dialog

The Dialog Control Language exposes a tile tree built from a .dcl file. Out of scope for the headless-first phase; tracked here so the data shape is visible (tile, action, list, image controls).

The Host Abstraction Layer (HAL)

Goal

Decouple the **what** (what the AutoLISP builtin does) from the **where** (which CAD backend stores or executes its effect).

Shape

A small set of generic functions per data domain. Each backend provides a method on every operation it supports; unsupported operations signal a uniform `:host-not-supported` runtime error.

```
(defgeneric host-entget        (host ename))
(defgeneric host-entmod        (host ename group-code-list))
(defgeneric host-entmake       (host group-code-list))
(defgeneric host-entdel        (host ename))
(defgeneric host-entlast       (host))
(defgeneric host-entnext       (host ename))

(defgeneric host-ssget         (host filter &key mode))
(defgeneric host-ssadd         (host pickset ename))
(defgeneric host-ssname        (host pickset index))
(defgeneric host-sslength      (host pickset))

(defgeneric host-tblsearch     (host table-kind name))
(defgeneric host-tblnext       (host table-kind &key rewind))

(defgeneric host-getvar        (host name))
(defgeneric host-setvar        (host name value))

(defgeneric host-command       (host argv))   ; the (command ...) builtin
(defgeneric host-prompt        (host string))
(defgeneric host-getstring     (host prompt &key controls))
(defgeneric host-getpoint      (host prompt &key base controls))
(defgeneric host-getreal       (host prompt &key controls))
(defgeneric host-getint        (host prompt &key controls))

(defgeneric host-grdraw        (host from to colour highlight))
(defgeneric host-grtext        (host kind ...))
(defgeneric host-grclear       (host))
```

Per-session backend

The `runtime-session` struct (autolisp-runtime/source/model.lisp) gains a slot:

```
(defstruct runtime-session
```

```
...
(host nil))
```

A new constructor argument `:host` defaults to a **NullHost** singleton that errors on every operation; production callers pass a MockHost or, eventually, a LiveHost.

Error codes

Existing runtime error codes (`:invalid-list-argument`, etc.) extend with:

- `:host-not-supported` — backend does not implement this operation
- `:host-not-connected` — operation requires a live CAD process
- `:host-protocol-error` — backend returned a malformed reply

Backends

NullHost

Trivial. Every generic function method signals `:host-not-supported`. Default for the strict and other test sessions; preserves current behaviour.

MockHost

In-memory deterministic CAD-database substitute. Owns:

- one `entity-table` (hash-table from ename id → entity-handle)
- one `pickset-table`
- one `vla-object-table`
- per-kind ~symbol-table~s for blocks, layers, ltypes, styles, views, ucs, vports, dimstyles, appids
- one `named-object-dictionary`
- one `sysvar-table` pre-populated with conservative defaults
- one `prompt-stream` (input source for `getstring` etc.)
- one `display-log` (records `grdraw` / `grtext` calls for inspection from tests)

The MockHost is the production runtime substrate for every clautolisp scenario that does not have a live CAD process behind it. It must be deterministic: the same input script under the same dialect, the same MockHost initial state, and the same `prompt-stream` must produce the same output and the same final database state.

GuiHost (optional, future)

A standalone Qt or web view that talks to clautolisp through the HAL via a JSON / S-expression protocol. Headless-test-friendly: the GUI is an observer of MockHost state, not a substitute for it.

LiveHost

A bridge to BricsCAD or AutoCAD. Speculative for now. Two plausible shapes:

- **Stdio bridge** — clautolisp drives an interactive AutoLISP REPL inside a real CAD process and reads back tagged S-expressions. Closest to the existing `autolisp-script` harness used for spec probes.
- **COM bridge** — Windows-only; the runtime drives the CAD process through its OLE Automation interface. More direct but Windows-bound.

Either way, LiveHost reuses the HAL contract: existing AutoLISP source runs unchanged when the backend is swapped.

Phased Roadmap

Numbering continues from Phase 6 (the Lisp-1, dynamic-scope, and general-semantics work that closed in 2026-04-26).

Phase 7 — Function-Coverage Round-Out

Goal: implement every host-independent builtin so that pure-data AutoLISP source loads cleanly. No HAL changes; the existing NullHost remains the default.

Scope (drives by `autolisp-spec` chapters 4-7, 9-13):

- Numeric / math: =sqrt, exp, log, log10, sin, cos, tan, asin, acos, atan, atan2 form, expt, mod, abs (already), random, floor, ceiling, round, fix (already)=
- Bitwise: `boole, ~ (already), logand (already), logior (already), logxor, lsh (already)`
- List: =append (already), reverse (already), nth (already), member (already), assoc (already), last (already), length (already), list (already), cons (already), subst (already), vl-list-length, vl-position, vl-remove (already), vl-sort, vl-sort-i, distance, remove= and the smaller corners.
- String: family already substantially in place; round out `vl-string-elt`, `vl-string-translate`, `vl-string-position` (done), `snvalid`, `wcmatch`, `xstrcase`.
- Predicate: `atom` (done), `numberp` (done), `zerop` (done), `minusp` (done), `listp` (done), `vl-consp` (done), `vl-symbolp` (done), `atoms-family`.
- Symbol: `boundp` (done), `vl-symbol-name` (done), `vl-symbol-value` (done), `set` (done), `atoms-family` (todo).
- I/O: `princ`, `prin1`, `print`, `terpri`, `read-line`, `read-char`, `write-line`, `write-char`, `open`, `close` (all done).
- Error / control: `vl-catch-all-apply` (done), `exit` / `quit` (done), `*error*` hook (done), `setfunhelp`, `vl-bt`, `vl-bt-on`, `vl-bt-off`.

Deliverables:

- one or two new sections in `autolisp-builtins-core/source/api.lisp`

- one new test file per family in `autolisp-builtins-core/tests/`
- coverage table (autogenerated from `core-builtins` vs the spec entries) in this document

Phase 8 — HAL Foundation [DONE 2026-04-26]

Goal: introduce the abstract host interface and the NullHost default. No behaviour change; a structural step.

Status: closed in commit **{phase-8}**. Suite 818 / 0; smoke-test of the existing real-world workflow runs unchanged.

Scope delivered:

- New ASDF system `clautolisp/autolisp-host` depending on `clautolisp/autolisp-runtime`. Three source files: `package.lisp`, `protocol.lisp` (every generic-function signature), `null-host.lisp` (the NullHost class + every method).
- Generic-function surface covers entity (`entget`, `entmod`, `entmake`, `entdel`, `entlast`, `entnext`, `handent`), selection set (`ssget`, `ssadd`, `ssdel`, `ssname`, `sslength`, `ssmemb`, `ssgetfirst`, `sssetfirst`), tables (`tblsearch`, `tblnext`, `tblobjname`), dictionaries (`namedobjdict`, `dictsearch`, `dictnext`, `dictadd`, `dictremove`, `dictrename`), sysvars (`getvar`, `setvar`), command dispatch (`command`), prompts (`prompt`, `initget`, `getstring`, `getint`, `getreal`, `getpoint`, `getcorner`, `getdist`, `getangle`, `getorient`, `getkword`), and graphics (`grdraw`, `grtext`, `grvecs`, `grclear`, `grread`, `redraw`).
- Runtime: `runtime-session` gains a `host` slot; new accessors `runtime-session-host`, `set-runtime-session-host`, `current-evaluation-host`. The `*default-runtime-host*` parameter holds the process-wide fallback; the autolisp-host module installs the NullHost singleton there at load time.
- New runtime error code: `:host-not-supported` (signalled by every NullHost method, with the backend name and operation name embedded in the message).
- CLI: new `--host NAME` flag, accepted values `null` (default), with `mock` reserved for Phase 9 (errors with a clear message). The REPL banner now reads "`clautolisp X.Y.Z - REPL (DIALECT dialect, HOST host)`". Builtins installation runs after the host is attached so future host-aware builtins see the right backend.
- Tests: 14 new tests in a fresh `clautolisp/autolisp-host/tests` system pin every NullHost method's error code, the default installation, the per-session keyword override, and `set-runtime-session-host`. Aggregate suite 818 / 0.

Note: no existing core builtin currently calls out to the host layer, so the "re-route" step listed in the original Phase 8 scope was a no-op. Phase 9 will introduce the first builtins that actually use the HAL.

Phase 9 — MockHost Data Structures [DONE 2026-04-26]

Goal: the in-memory substrate for the entity / pickset / table / sysvar layers. No host operations yet; only the data carriers.

Status: closed. Suite 883 / 0; `clautolisp --host mock` now constructs a usable MockHost (host operations still error with `:host-not-supported` until Phase 10 fills in the methods).

Scope delivered:

- New ASDF system `clautolisp/autolisp-mock-host` depending on `clautolisp/autolisp-host` and `clautolisp/autolisp-runtime`. Four source files: `package.lisp`, `model.lisp` (struct defs + the `mock-host` class), `sysvars.lisp` (default tables and sysvars), `api.lisp` (constructor + snapshot/restore).
- Structs delivered as sketched in the data-model section: `entity-handle`, `pickset`, `symbol-table-record`, `dictionary`, `sysvar-cell`, plus the `mock-host` class itself with slots for entities, picksets, vla-objects, per-kind table hash, named-object dictionary, sysvars, prompt-stream, prompt-output, display-log, pickfirst.
- Pre-populated symbol tables: `:block-record` holds `"*Model_Space"` / `"*Paper_Space"`; `:layer` holds `"0"`; `:ltype` holds `"BYBLOCK"` / `"BYLAYER"` / `"Continuous"`; `:style` and `:dimstyle` each hold `"Standard"`; `:vport` holds `"*Active"`; `:appid` holds `"ACAD"`. `:ucs` and `:view` start empty.
- Pre-populated sysvars (CMDECHO, CECOLOR, CLAYER, ANGBASE, ANGDIR, AUNITS, OSMODE; USERR1..R5, USERI1..I5, USERS1..S5; plus read-only DWGNAME / DWGPREFIX / PLATFORM and writable LISPSYS). Lookup is case-insensitive; writes to read-only cells signal `:host-not-supported`.
- `mock-host-snapshot` / `mock-host-restore` pair: the snapshot is a property list of entities, picksets, tables, sysvars, and pickfirst pointer; round-trip restore preserves the host's AutoLISP-visible state. Tests pin determinism.
- Convenience accessors: `mock-host-table`, `mock-host-find-table-record`, `mock-host-add-table-record`, `mock-host-sysvar`, `mock-host-set-sysvar`.
- Base-class fallback methods on `host` for every HAL operation signalling `:host-not-supported`. This lets MockHost (with no method overrides yet) inherit a sensible "not yet implemented" failure shape; Phase 10 will override the methods that MockHost actually serves.
- CLI: `--host mock` now constructs a fresh MockHost (instead of erroring as a Phase-9-todo). The REPL banner reads "...REPL (DIALECT dialect, mock-host host)~~~". Host operations still error until Phase 10.
- Tests: 14 new tests in `clautolisp/autolisp-mock-host/tests` cover class hierarchy, empty fixture, default tables, default sysvars, sysvar mutation including read-only refusal, case-insensitive lookup, snapshot/restore round-trip, slot shapes for prompt-output / display-log / pickset / dictio-

nary, and the inherited `:host-not-supported` behaviour. Aggregate suite 76 + 184 + 29 + 52 + 412 + 130 = 883 / 0.

Phase 10 — MockHost Entity API [DONE 2026-04-26]

Goal: implement the entity-level HAL methods against MockHost.

Status: closed. Suite 919 / 0; the AutoLISP-visible entity round trip (entmake -> entget / entlast / entnext / handent / entmod / entdel) runs end-to-end against `--host mock`.

Scope delivered:

- New file `clautolisp/autolisp-mock-host/source/entity-api.lisp` defines methods on `mock-host` for `host-entget`, `host-entmod`, `host-entmake`, `host-entmakex`, `host-entdel` (with the documented within-command toggle), `host-entupd`, `host-entlast`, `host-entnext` (both zero-arg and one-arg forms), `host-handent`.
- `mock-host` class extended with a hex-string handle allocator (`mock-host-allocate-handle`), a `creation-order` list (walked by entlast / entnext), and an `entities` hash-table re-keyed on the handle string.
- `entmake` / `entmakex` accept a DXF group-code list with a `(0 . "TYPE")` marker; the host injects `(-1 . ENAME)` at the head and `(5 . HANDLE)` as the second entry, mirroring AutoCAD's on-disk shape, and stores the resulting list. Missing marker signals `:invalid-entity-data`.
- `entget` on a deleted entity returns nil; `entdel` toggles the `deleted-p` flag (re-entering "live" on a second call within the same scope, as documented).
- `handent` accepts both CL strings and AutoLISP-strings, returns an ENAME for known handles and nil otherwise.
- `clautolisp/autolisp-builtins-core` now depends on `clautolisp/autolisp-host`. New AutoLISP builtins `ENTGET`, `ENTMOD`, `ENTMAKE`, `ENTMAKEX`, `ENTDEL`, `ENTUPD`, `ENTLAST`, `ENTNEXT`, `HANDENT` are thin wrappers over the HAL generic functions, dispatched on the active session's host backend.
- `autolisp-runtime` now exports `make-autolisp-ename` (the public ENAME constructor used by the HAL).
- The `autolisp-value->string` printer renders ENAMEs as "<Entity name: HANDLE>" so the REPL banner displays them cleanly under `--host mock`.
- snvalid at the host level (cross-checking against current table-record names) is parked for Phase 11 alongside tblsearch / tblnext, since it relies on table walkers.

Tests:

- 11 new entity-api tests in `autolisp-mock-host/tests/entity-api-tests.lisp` pin every method's contract: handle allocation, bookkeeping injection, round-trip via entget, delete-toggle, entmod-replace, creation-order walk

under entlast / entnext, deleted-skip, handent lookup, invalid-data error path, invalid-ename error path.
- 1 new builtin-level test in autolisp-builtins-core/tests pins the AutoLISP-side wiring under both NullHost and MockHost.
- Aggregate suite 76 + 184 + 29 + 82 + 418 + 130 = 919 / 0.

Smoke: `clautolisp --host mock -x "(setq e (entmake ...)) (entget (cdr (assoc -1 e)))"` runs the entmake -> entget round trip end-to-end and prints the layer name from the stored data list.

Phase 11 — MockHost Selection / Tables / Sysvars [DONE 2026-04-26]

Goal: selection sets, table walkers, and system variable access on MockHost.

Status: closed. Suite 959 / 0; the AutoLISP-side ‘(ssget "X" filter)‘ / ‘(tblsearch . . .)‘ / ‘(getvar . . .)‘ / ‘(setvar . . .)‘ flow runs end-to-end against `--host mock`.

Scope delivered:

- New file `autolisp-mock-host/source/selection-api.lisp` implements `host-ssget` (modes "X" / "_X" with optional filter-list; other modes signal :unsupported-ssget-mode), `host-ssadd` (zero / one / two arg shapes), `host-ssdel`, `host-ssname` (zero-based), `host-sslength`, `host-ssmemb`, `host-ssgetfirst`, `host-sssetfirst`. Filter-list matching honours comma-separated alternation and the ‘*‘ / ‘¿ wildcards on string values. Group-code -4 logical operators (<AND, <OR, <NOT) are deferred.
- New file `autolisp-mock-host/source/table-api.lisp` implements `host-tblsearch`, `host-tblnext` (with per-kind iterator state on `mock-host-tblnext-iterators`; rewind restarts), and `host-tblobjname` returning an ENAME wrapping the table-record's internal id.
- New file `autolisp-mock-host/source/sysvar-api.lisp` implements `host-getvar` (wraps strings in autolisp-string for AutoLISP consumption) and `host-setvar` (per-kind coercion: integers truncate reals; reals coerce numbers; strings reject non-strings; points validate 2D/3D coordinate lists). Read-only cells signal :sysvar-read-only; unknown names signal :unknown-sysvar.
- `autolisp-builtins-core` gained 13 new AutoLISP builtins routing through the HAL: `SSGET`, `SSADD`, `SSDEL`, `SSNAME`, `SSLENGTH`, `SSMEMB`, `SSGETFIRST`, `SSSETFIRST`, `TBLSEARCH`, `TBLNEXT`, `TBLOBJNAME`, `GETVAR`, `SETVAR`.
- Runtime exports new constructor `make-autolisp-pickset`.

Tests:

- 21 new tests in `autolisp-mock-host/tests/selection-tests.lisp` pin: ssget X / X+filter / type / layer / comma-alternation / wildcard / empty result / no-mode / unsupported-mode; ssadd / ssdel / ssname positional / ssmemb membership / ssgetfirst+sssetfirst round-trip; tblsearch hit + miss;

tblnext walk and rewind; tblobjname returns ename; getvar with string wrapping; getvar on unknown name returns nil; setvar coerces integer / real / string; setvar rejects read-only / unknown / type-mismatch.
- Aggregate suite 76 + 184 + 29 + 122 + 418 + 130 = 959 / 0.

Smoke (from the CLI under `--host mock`):

```
clautolisp --host mock -x \
  "(princ (getvar \"CLAYER\"))            ; -> 0
   (setvar \"CLAYER\" \"Drafting\")
   (princ (getvar \"CLAYER\"))            ; -> Drafting
   (princ (cdr (assoc 2 (tblsearch \"LAYER\" \"0\"))))  ; -> 0
   (entmake (list (cons 0 \"LINE\") (cons 8 \"0\")))
   (entmake (list (cons 0 \"CIRCLE\") (cons 8 \"0\")))
   (princ (sslength (ssget \"X\" (list (cons 0 \"LINE\")))))"  ; -> 1
```

Phase 12 — Headless Interaction Channel

Goal: let real AutoLISP scripts that read user input run in batch.

Scope:

- `getstring`, `getreal`, `getint`, `getpoint`, `getcorner`, `getdist`, `getangle`, `getorient`, `getkword` — all read from a user-supplied stream attached to MockHost.
- `initget` — input-control flags applied to subsequent `get` calls.
- `prompt`, `princ` — write to MockHost's prompt stream.
- `grread` — minimal headless model (immediate eof / configurable scripted events).

The CLI gains a flag `--mock-input PATH` pointing at a script-driven response file, so a test rig can feed deterministic answers.

Phase 13 — VLA / VLAX Foundations

Goal: the COM compatibility surface, headless-first.

Scope (drives by `autolisp-spec` chapter 19):

- `vlax-create-object`, `vlax-get-object`, `vlax-release-object` (against a MockHost-managed COM-object table).
- `vlax-get-property`, `vlax-put-property`, `vlax-invoke-method` with documented method-name mapping.
- `vlax-make-safearray`, `vlax-safearray-fill`, `vlax-safearray-get-element`, `vlax-safearray-put-element`, `vlax-safearray->list`.
- `vlax-make-variant`, `vlax-variant-type`, `vlax-variant-value`, `vlax-variant-change-type`.
- `vlax-import-type-library` — implementation-defined; declarative description of mock-COM type libraries.

- `vla-*` wrappers — generated from a small declarative table of COM dispatch.

Phase 14 — Reactors

Goal: the Visual LISP reactor surface.

Scope (`autolisp-spec` chapter 22):

- All `vlr-*` functions: registration, callback dispatch, owner bookkeeping, persistence on the named-object dictionary.
- A MockHost-side event injection helper for tests (`mock-host-fire-event :command-will-start ...`).

Phase 15 — DCL and Optional GUI

Goal: dialog support, optional GUI backend.

Scope:

- DCL parser + tile model (`load_dialog`, `new_dialog`, `start_dialog`, `done_dialog`, `unload_dialog`).
- `action_tile`, `set_tile`, `get_tile`, `mode_tile`, `start_image`, `vector_image` etc.
- Headless backend (records tile actions, no rendering).
- Optional GUI backend (Qt or web) — separate worktree, opt-in.

Phase 16 — LiveHost Adapter

Goal: run AutoLISP scripts through clautolisp against an actual BricsCAD or AutoCAD process.

Scope (highly speculative):

- Stdio-bridge prototype against BricsCAD (Linux / macOS are feasible without ActiveX).
- COM bridge for AutoCAD on Windows.
- Integration with the existing `autolisp-spec` probe harness.

Cross-Cutting Concerns

Coverage tracking

A new tooling target `make builtins-coverage` produces a report comparing `core-builtins` entries against the function inventory in `autolisp-spec/documentation/autolisp-visual-lisp-specification-draft.org`, broken down by chapter. Each phase above closes a column of that report.

Testing strategy

- **Unit tests** (FiveAM) per builtin family.
- **Snapshot tests** against MockHost: a script runs, the post-condition entity table is dumped, and the dump is compared against a stored `.snapshot` file.
- **Probe replay** against the BricsCAD V26 / macOS harness for semantics that are not directly testable headlessly (used at spec-time, not at every build).
- **File-compat scenarios** (`autolisp-file-compat`) continue to function as a declarative scenario corpus.

Compatibility profile vs dialect

The dialect descriptor (`strict`, `autocad-2026`, `bricscad-v26`) describes the **language**. The HAL backend describes the **host**. They are independent: a script under the `autocad-2026` dialect can run against MockHost; a script under `strict` can run against LiveHost. Tests must exercise the cross product.

Versioning

`clautolisp/tools/clautolisp/source/version.lisp` bumps DEVELOP on every source change. Phase boundaries optionally bump MINOR (0.0.x → 0.1.0 → ...). MAJOR is reserved for API-breaking changes (e.g., a HAL contract revision).

Open Architectural Questions

These are recorded here so they are remembered when their phase arrives:

1. **ENAME identity across sessions.** If LiveHost reconnects, are ENAMEs from a previous session re-resolvable? Probably no: every session gets a fresh ID space.
2. **Concurrent sessions.** Today's runtime is single-session. Is a pool of MockHosts useful for parallel test execution? Likely yes — `runtime-session` already isolates sessions; the open work is making MockHost itself thread-safe.
3. **Garbage collection of pickset / VLA tables.** A long-running session leaks. Provide an explicit `clear-host-handles` helper on MockHost; let LiveHost translate it to `object-released-p` sweeps.
4. **`(command ...)` interactivity in headless mode.** The `command` builtin in real AutoCAD drives the command line. In MockHost we record commands to a log; `initget` / `getstring`-like prompts feeding into `command` need a coherent contract.
5. **DCL rendering.** Headless DCL records actions. A production GUI backend will need to choose between Qt, web, or platform-native; the HAL

contract should be neutral on that question.

6. **COM type library scope.** A real AutoLISP corpus references a long tail of `vla-*` functions generated from imported type libraries. We will have to draw a line between "documented" and "user-imported" and only ship the documented surface in core.

Cross-References

- **Spec**: `autolisp-spec/documentation/autolisp-visual-lisp-specification-draft.org` — chapter index drives every phase.
- **Existing PLAN**: `clautolisp/PLAN.md` — actionable tasks. This roadmap is the parent document.
- **Sibling design notes**: `clautolisp/documentation/development-plan.org` — earlier high-level architecture rationale.
- **User manual**: `clautolisp/documentation/user-manual.org` — the CLI surface, kept in sync with `tools/clautolisp/source/main.lisp`.